

NOETHER: Data and Artifact Availability Statement

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Data and Artifact Availability

To support independent verification, the supplementary code and data are archived on Zenodo under DOI <https://doi.org/10.5281/zenodo.20250634>. The same materials are supplied through the submission system with the same file manifest and content hash.

Review-stage supplementary archive. The following supplementary materials are submitted alongside the manuscript and are available through the submission system under SHA-256 content hash. The hash is computed over the concatenated tar-archive of the listed items and reported in the final manuscript at acceptance.

- **S1** Python reference implementation of CONSTRUCT-MP described in supplementary S9, including the Boltzmann and equivariant-ML instantiations.
- **S2** The full PWR-MR corpus, with each MR annotated by canonical structural component, NOETHER MetaPattern, and source equation.
- **S3** The SE(3)-equivariant point-cloud testing harness, including the mutation generators and MR sets used in the case-study comparison.

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- **S4** A reproducibility manifest documenting the random seeds, model checkpoints, and dataset versions referenced in the case study.
- **S5** GenMorph/PIT pilot material, head-to-head summaries, parsed mutation results, and statistical scripts for the secondary executability and complementarity analyses.
- **S6** Query-optimiser pilot material for the relational-equivalence extension, including sampled Calcite/QED plan pairs and oracle scripts.
- **S7** Defects4J algebra-rich substrate material, pre-registration records, SUT-selection criteria, and $\mathcal{L}^*$ -blindness evidence.
- **S8** METRIC+ comparison material, Sun et al. subject re-implementations, scope analysis, and head-to-head protocol/results files.
- **S9** Appendices and detailed empirical tables migrated out of the Fast Impact main body for length conformance.
- **S10** Home-field generation-to-detection checks, rank/faithfulness scripts, and additional Invariance-Blindness sanity checks.
- **S11** Industrial N5 coverage classification inputs and results.
- **S12** Cross-domain N5 numerical-library and signal-processing derivation checks and outputs.

Public archival release. The public archival package is deposited on Zenodo under DOI <https://doi.org/10.5281/zenodo.20250634>. The camera-ready version will additionally record the final SHA-256 hash of the submission archive. We aim for the *Available* and *Functional* artefact-evaluation badges.